

Eigenproblems and Physical Modes

An eigenproblem becomes physically meaningful when its eigenvalues, eigenvectors, units, and validation checks are connected back to measurable collective motion.

Physical question. How can two coupled masses move in special collective patterns that preserve their relative shape while oscillating at characteristic frequencies?

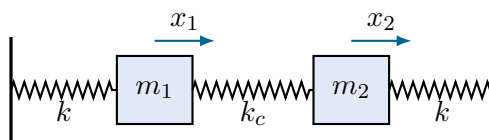
Learning outcomes

By the end of this week, you should be able to:

1. derive a two-degree-of-freedom oscillator model from Newton's second law;
2. formulate the generalized eigenproblem $\mathbf{K}\phi = \lambda\mathbf{M}\phi$;
3. connect λ to angular frequency and frequency with correct units;
4. normalize and interpret eigenvectors as physical mode shapes without over-interpreting arbitrary scale or sign;
5. validate computed modes using symmetry, analytic references, eigen-equation residuals, orthogonality, and limiting cases; and
6. identify one preferred bounded capstone question and one alternate candidate with feasible validation evidence.

1. From simultaneous motion to special patterns

Coupled systems have several coordinates that influence one another. In a two-mass oscillator, a displacement of mass 1 changes the force on mass 2 through the coupling spring, and vice versa. Most initial conditions produce a mixture of motions, but some displacement patterns are special: the entire shape simply scales up and down while oscillating at one characteristic frequency. These are the **normal modes**.



Predict before computing. For two equal masses and identical outer springs, sketch one mode where the masses move together and one where they move oppositely. Which should have the larger frequency? Explain using deformation of the coupling spring.

2. Derive the equations of motion

Let both masses equal m , the two outer springs equal k , and the coupling spring equal k_c . Measure x_1 and x_2 from static equilibrium and take rightward displacement as positive. For small undamped motion,

$$m\ddot{x}_1 = -(k + k_c)x_1 + k_cx_2, \quad (1)$$

$$m\ddot{x}_2 = k_cx_1 - (k + k_c)x_2. \quad (2)$$

Move the restoring terms to the left and collect the coordinates:

$$\underbrace{\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}}_{\mathbf{M} \text{ (kg)}} \underbrace{\begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix}}_{\ddot{\mathbf{x}} \text{ (m s}^{-2}\text{)}} + \underbrace{\begin{bmatrix} k + k_c & -k_c \\ -k_c & k + k_c \end{bmatrix}}_{\mathbf{K} \text{ (N m}^{-1}\text{)}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbf{x} \text{ (m)}} = 0. \quad (3)$$

The off-diagonal terms encode the coupling. Their negative sign is not a MATLAB convention; it comes from the restoring force created by relative displacement.

3. Harmonic motion turns the model into an eigenproblem

For one normal mode, assume

$$\mathbf{x}(t) = \phi \cos(\omega t), \quad (4)$$

where ϕ is a fixed displacement pattern. Then

$$\ddot{\mathbf{x}}(t) = -\omega^2 \phi \cos(\omega t). \quad (5)$$

Substitution into $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0$ gives

$$(\mathbf{K} - \omega^2 \mathbf{M})\phi = 0, \quad (6)$$

or equivalently

$$\boxed{\mathbf{K}\phi = \lambda \mathbf{M}\phi}, \quad \lambda = \omega^2. \quad (7)$$

This is a **generalized eigenproblem**. Since stiffness divided by mass has units $(\text{N m}^{-1})/\text{kg} = \text{s}^{-2}$, the eigenvalue has units s^{-2} . Therefore

$$\omega = \sqrt{\lambda} \quad (\text{rad s}^{-1}), \quad f = \frac{\omega}{2\pi} \quad (\text{Hz}). \quad (8)$$

Common interpretation error. The eigenvalue is not the frequency in hertz. For this mechanical formulation, the eigenvalue is ω^2 . Taking a square root and then dividing by 2π is part of the physical interpretation, not cosmetic unit conversion.

4. Baseline symmetric system

Use

$$m = 0.50 \text{ kg}, \quad k = 20 \text{ N m}^{-1}, \quad k_c = 10 \text{ N m}^{-1}. \quad (9)$$

The generalized eigenproblem can be solved directly in MATLAB:

Listing 1: Generalized eigenproblem for the two-mass oscillator

```
1 M_kg = diag([0.50 0.50]);
2 K_Npm = [30 -10; -10 30];
3
4 [modeRaw,D] = eig(K_Npm,M_kg);
5 [lambda_per_s2,idx] = sort(diag(D));
6 modeRaw = modeRaw(:,idx);
7
8 omega_radps = sqrt(lambda_per_s2);
9 frequency_Hz = omega_radps/(2*pi);
```

The reference values are

$$\lambda_1 = 40 \text{ s}^{-2}, \quad \lambda_2 = 80 \text{ s}^{-2}, \quad (10)$$

$$\omega_1 = 6.32456 \text{ rad s}^{-1}, \quad \omega_2 = 8.94427 \text{ rad s}^{-1}, \quad (11)$$

and

$$f_1 = 1.00658 \text{ Hz}, \quad f_2 = 1.42353 \text{ Hz}. \quad (12)$$

5. Eigenvectors are relative mode shapes

For the symmetric system, the two physical patterns are proportional to

$$\phi_1 \propto \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \phi_2 \propto \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (13)$$

Mode 1 is **in phase**: both masses move together, so the coupling spring does not change length. Its frequency therefore depends only on k/m . Mode 2 is **out of phase**: the coupling spring is repeatedly stretched and compressed, increasing the effective restoring action and therefore the frequency.

An eigenvector has arbitrary nonzero scale. MATLAB may return $[-1, -1]^T$ instead of $[1, 1]^T$; these represent the same physical mode. Normalize before comparing shapes, for example by setting the largest absolute component to one. The sign may also be reversed for consistent display.

6. Symmetry provides an analytic validation case

Substitute the expected symmetry patterns directly into the equations. For the in-phase mode, $x_1 = x_2$ and the coupling spring is unstretched:

$$\omega_1^2 = \frac{k}{m} = 40 \text{ s}^{-2}. \quad (14)$$

For the out-of-phase mode, $x_2 = -x_1$:

$$\omega_2^2 = \frac{k + 2k_c}{m} = 80 \text{ s}^{-2}. \quad (15)$$

These results exactly match the numerical eigenvalues for the baseline parameters. This is stronger evidence than merely observing that MATLAB completed without an error.

7. Residual and orthogonality answer different questions

For each computed eigenpair, form the residual

$$\mathbf{r}_i = \mathbf{K}\phi_i - \lambda_i \mathbf{M}\phi_i. \quad (16)$$

A useful scaled residual is

$$\rho_i = \frac{\|\mathbf{r}_i\|_2}{\|\mathbf{K}\|_2 \|\phi_i\|_2 + |\lambda_i| \|\mathbf{M}\|_2 \|\phi_i\|_2}. \quad (17)$$

This tests whether the eigenpair satisfies the assembled matrices. It does *not* prove that the matrices were derived correctly.

For a symmetric conservative system with distinct modes, mass-normalized eigenvectors satisfy

$$\Phi^T \mathbf{M} \Phi \approx \mathbf{I}. \quad (18)$$

This gives a structurally different check from the eigen-equation residual.

Evidence	Question answered	Important limitation
Symmetry / analytic result	Does the model reproduce a known special case?	Applies only when its assumptions hold
Eigen residual	Does the eigenpair satisfy the assembled matrices?	Can be tiny for the wrong matrices
Mass orthogonality	Are distinct conservative modes structurally consistent?	Depends on model symmetry and definiteness
Limiting case	Does the result approach known behaviour as a parameter changes?	Requires a trustworthy limit

8. Coupling strength creates frequency splitting

For the symmetric model,

$$\omega_1^2 = \frac{k}{m}, \quad \omega_2^2 = \frac{k + 2k_c}{m}. \quad (19)$$

Increasing k_c therefore leaves the in-phase frequency unchanged while increasing the out-of-phase frequency. The separation between the two natural frequencies is a measurable signature of coupling.

The limiting case $k_c \rightarrow 0$ is especially useful. Then the masses become independent and the two eigenvalues are equal:

$$\lambda_1 = \lambda_2 = \frac{k}{m}. \quad (20)$$

This repeated eigenvalue is called a **degeneracy**. At degeneracy, individual eigenvectors are not unique: any independent basis spanning the same two-dimensional eigenspace is acceptable. Therefore, demanding one exact pair of vectors from MATLAB is not a valid test.

9. Breaking symmetry changes the mode shapes

Now keep $m_1 = 0.50$ kg but increase m_2 to 0.75 kg while retaining $k = 20$ N m⁻¹ and $k_c = 10$ N m⁻¹. The frequencies become approximately

$$f_1 = 0.88401 \text{ Hz}, \quad f_2 = 1.32346 \text{ Hz}. \quad (21)$$

The mode shapes are no longer exact $[1, 1]^T$ and $[1, -1]^T$ patterns. Unequal inertia changes the relative amplitudes, so the simple symmetric analytic formula no longer applies. Use the generalized eigen-equation residual, mass orthogonality, positive squared frequencies, and physically reasonable parameter trends instead.

Responsible AI use. AI may help translate derived equations into MATLAB, but it can confuse λ , ω , and f , mishandle eigenvector sign or normalization, and overstate residual evidence. Record accepted, modified, or rejected suggestions and the independent checks used.

10. Practical transfer and capstone progression

The practical starts from the symmetric baseline and makes two controlled changes:

1. vary k_c to observe frequency splitting and the zero-coupling degeneracy; and
2. increase one mass to break symmetry and interpret the new mode shapes.

For each change, predict a trend, compute evidence, validate it, and explain the physics rather than collecting numbers without a purpose.

By the end of Week 4, nominate one preferred capstone question and one alternate. State the physical system, controllable input, observable output, likely method, required data or parameters, and at least two independent checks. The choice remains provisional until Week 06.

Three takeaways

1. The generalized eigenproblem comes from the equations of motion; $\lambda = \omega^2$, not frequency in hertz.
2. Eigenvectors describe relative mode shape, so overall scale and sign are arbitrary.
3. Trust requires layered evidence from analytic cases, residuals, orthogonality, and limiting behaviour.

Preparation for the practical. Complete the Week 04 Individual Practical Check — Quiz in Google Classroom. Bring your derived two-mass equations and be ready to solve, validate, vary coupling, break symmetry, and record two capstone candidates.